Calculus I Assignment 10

1. Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of the function.

   (a) $y = \int_0^{\tan x} e^{-t^2} dt$

   (b) $y = \int_{\sqrt{x}}^{\pi/4} \theta \tan\theta d\theta$

2. Evaluate the integral.

   (a) $\int_1^4 \frac{2+x^2}{\sqrt{x}} dx$

   (b) $\int_0^{\pi/3} \sec\theta \tan\theta d\theta$

   (c) $\int_0^1 (1+r)^3 dr$

   (d) $\int_{1/2}^{1/\sqrt{2}} \frac{4}{\sqrt{1-x^2}} dx$

3. On what interval is the function

$$y = \int_0^x \frac{t^2}{t^2+t+2} dt$$

   concave?

4. If $f(x) = \int_0^{\sin x} \sqrt{1+t^2} dt$ and $g(y) = \int_3^y f(x) dx$, find $g''(\pi/6)$.

5. Use l'Hospital's Rule to evaluate the following limit.

$$\lim_{x\to\infty} \frac{1}{x^2} \int_0^x \ln\left(1+e^t\right) dt.$$

6. Evaluate the following limit by first recognizing the sum as a Riemann sum for a function defined on $[0,1]$:.

$$\lim_{n\to\infty} \frac{1}{n} \left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \sqrt{\frac{3}{n}} + \cdots + \sqrt{\frac{n}{n}} \right).$$

7. Evaluate the definite integral.

   (a) $\int_1^4 \frac{\sqrt{2+\sqrt{x}}}{\sqrt{x}} dx$

   (b) $\int_0^a x\sqrt{x^2+a^2} dx \quad (a > 0)$

   (c) $\int_1^{16} \frac{x^{1/2}}{1+x^{3/4}} dx$

8. Determine whether the integral is convergent or divergent. Evaluate integrals that are convergent.

   (a) $\int_1^\infty \frac{x^2+x+1}{x^4} dx$

   (b) $\int_2^\infty \frac{x}{\sqrt{x^2-1}} dx$

   (c) $\int_{-\infty}^\infty x e^{-x^2} dx$

(d) $\int_2^\infty \frac{dv}{v^2+2v-3}$

(e) $\int_1^\infty \frac{dx}{\sqrt{x}+x\sqrt{x}}$

(f) $\int_0^{\pi/2} \tan^2\theta d\theta$

9. Use the Comparison Theorem to determine whether the integral is convergent or divergent.

(a) $\int_0^1 \frac{\sec^2 x}{x\sqrt{x}} dx$

(b) $\int_0^\pi \frac{\sin^2 x}{\sqrt{x}} dx$